



Hands-on Lab: Getting Started with Google Looker Studio

Estimated time needed: 60 minutes

Looker Studio, from Google, is a data discovery platform available to analyze and perform data-driven functionalities. Looker is known for its data exploration, visualization, and reporting capabilities. It empowers users to seamlessly connect with diverse data sources, enabling them to build interactive dashboards and generate insightful reports, thereby facilitating a comprehensive understanding of their data.

In this lab, you will learn how to sign up for Looker Studio and learn general navigation around the Looker user interface (UI). Next, you will learn how to upload external data files to Looker through connectors and then learn how to start a new dashboard with templates. Lastly, you will learn how to create a simple dashboard.

Dataset Used in this Lab

The dataset used in this lab is published by IBM. You can download the dataset file directly from here: [CustomerLoyaltyProgram.csv](#).

Objectives

After completing this lab, you will be able to:

- Sign up to use Looker Studio
- Navigate around the Looker Studio user interface
- Create a data source using a connector
- Access report themes and layouts
- Create a simple dashboard report

Exercise 1: Sign up for Looker Studio

In this exercise, you will learn how to sign up for Google's Looker Studio

1. Go To [Looker Studio](#)

Looker Studio Overview

STEP 1

https://lookerstudio.google.com/u/0/

Getting Started

Other Bookmarks

Looker Studio | Overview

OVERVIEW | GALLERY | CONNECT TO DATA | VISUALIZATIONS | HOME

Looker Studio

Your data is beautiful. Use it.

Unlock the power of your data with interactive dashboards and beautiful reports that inspire smarter business decisions. It's easy and free.

USE IT FOR FREE

STEP 2

https://datastudio.google.com/?requirelogin=1

2. Click **USE IT FOR FREE**.
3. A new window will open. If you already have a Google account, enter your credentials and click **Next** as shown below (number 1 and 2). Or click on **Create account** (number 3) and follow the steps.

The image shows a Google Sign in page for Looker Studio. It includes the Google logo, a 'Sign in' heading, and a subheading 'to continue to Looker Studio'. There is an input field for 'Email or phone' with a red arrow and the number '1' pointing to it. Below the input field is a link for 'Forgot email?'. Further down is a message about using a private browsing window with a 'Learn more' link. At the bottom, there are two buttons: 'Create account' (highlighted with a red box and the number '3') and 'Next' (highlighted with a red arrow and the number '2').

Google

Sign in

to continue to Looker Studio

1

Email or phone

Forgot email?

Not your computer? Use a private browsing window to sign in. [Learn more](#)

2

3

Create account

Next

Create

Recent

Reports

Data sources

Explorer

Recent

Start with a Template

Template Gallery

Shared with me

Owned by me

Trash

Templates

Blank Report

Tutorial Report

Acme Marketing

Search Console Report

Name	Owned by anyone	Last opened by me	Location
Untitled Report	P P	12:55 PM	Owned by me
Untitled Report	P P	Aug 31, 2023	Owned by me
Untitled Report	P P	Aug 31, 2023	Owned by me
Untitled Report	P P	Aug 30, 2023	Owned by me

Exercise 2: Navigate around the Looker Studio User Interface

In this exercise, you will understand Looker Studio UI components which you'll use further to create visuals and dashboards.

The goal of this exercise is to introduce you to the primary components and functionalities within Looker Studio.

On the home page of Looker Studio, you can conveniently create and access all your essential assets, including reports, data sources, and explorations.

The screenshot shows the Looker Studio interface. At the top, the 'Looker Studio' logo is on the left, a search bar labeled 'Search Looker Studio' is in the center, and user controls (help, settings, profile) are on the right. Below the header, a 'Create' button (1) is on the left. A 'Recent' section (2) contains tabs for 'Reports', 'Data sources', and 'Explorers'. The 'Start with a Template' section (6) features a 'Template Gallery' dropdown. Below this, four template cards are shown: 'Blank Report Looker Studio' (3), 'Tutorial Report Looker Studio' (7), 'Acme Marketing Google Analytics', and 'Search Console Report Search Console'. On the far left, a sidebar lists 'Shared with me', 'Owned by me', 'Trash', and 'Templates'. At the bottom, a table (4) lists recent assets.

Name	Owned by anyone	Last opened by me	Location
Report_Car_Sales		Aug 30, 2023	Owned by me
COVID_19_Dashboard_practice		Aug 30, 2023	Owned by me
BU COVID-19 Report		Aug 29, 2023	Shared with me

Let's understand the major components available on the homepage.

1. From here you can create a new asset such as a Report, a Data source or an Explorer.
2. This is where you access your recent Reports, Data sources, and Explorers.
3. With the Report tab selected, this is how you can start to create a blank report.
4. This lists any recently worked on assets. You can click the ellipsis button (...) next to an asset to perform actions on it, such as sharing, renaming, or removing it.

The screenshot shows the Looker Studio interface. At the top, there are four report thumbnails: 'Blank Report', 'Tutorial Report', 'Acme Marketing', and 'Search Co'. Below these is a table listing reports. A context menu is open over the 'Report_Car_Sales' row, showing options: 'Share', 'Rename', and 'Remove'. The 'Report_Car_Sales' row is highlighted with a red box around the menu icon.

Name	Owned by anyone	Last opened by me	Location
Report_Car_Sales	P P	Aug 30, 2023	Owned by me
COVID_19_Dashboard_practice	P P	Aug 30, 2023	Owned by me

5. Here you can search and find your Looker Studio assets quickly and the result will appear in the list at section 4.
6. You can choose a template from the Template Gallery to start creating an asset from.
7. Here you can take a tutorial on Looker Studio.

Exercise 3: Create a Data Source and Use Report Editor

Task 1: Create a data source

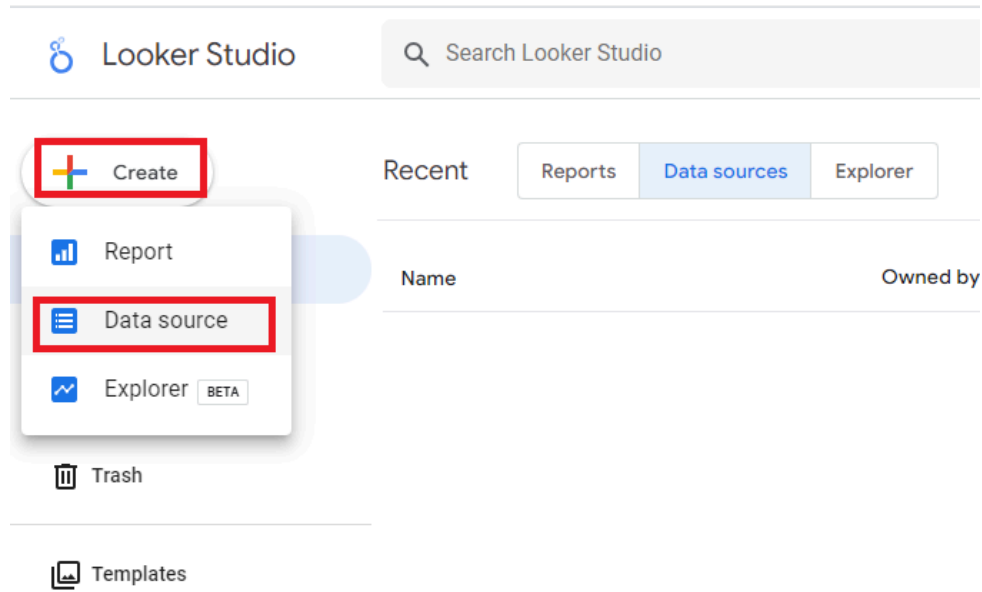
The first thing you need to start creating a report is to acquire some data.

To select an existing data source you would click the **Data sources** tab and your existing data sources will be listed.

The screenshot shows the Looker Studio interface. At the top left is the Looker Studio logo. To its right is a search bar labeled 'Search Looker Studio'. In the top right corner are icons for help, settings, and a user profile. Below the header, on the left, is a 'Create' button with a plus icon, highlighted with a red box and the number '2'. To the right of the 'Create' button are three tabs: 'Recent', 'Reports', and 'Data sources'. The 'Data sources' tab is highlighted with a red box and the number '1'. Below the tabs is a table with columns: 'Name', 'Owned by anyone', 'Last opened by me', and 'Location'. The table is currently empty. To the left of the table is a sidebar with a 'Recent' button and three categories: 'Shared with me', 'Owned by me', and 'Trash'. Below the sidebar is a 'Templates' button. In the center of the page, there is a large icon of a document with a list, and below it, the text 'Create a Data Source.' followed by the underlined text 'Use the Create button to add one.'

However, for this lab, you will create a new data source.

1. In the top left corner, click **Create**, then select **Data source**.



The new window that opens displays a lot of options for connecting to your data; these are called *Connectors*. A connector links Looker Studio to your data. Connecting to your data creates a data source within Looker Studio. Looker Studio provides a variety of connectors to connect to different kinds of data to create reports.

You can use the search field to look for the relevant data connector.



Google Connectors (23)

Connectors built and supported by Looker Studio [Learn more](#)

 Looker By Google Connect to your Looker semantic models.	 Google Analytics By Google Connect to Google Analytics.	 Google Ads By Google Connect to Google Ads performance report data.
 Google Sheets By Google Connect to Google Sheets.	 BigQuery By Google Connect to BigQuery tables and custom queries.	 AppSheet By Google Connect to AppSheet app data.
 File Upload By Google Connect to CSV/Excel/JSON data.	 Amazon Redshift By Google Connect to Amazon Redshift.	 Campaign Manager 360 By Google Connect to Campaign Manager 360.

Partner Connectors (839)

Connectors built and supported by Looker Studio partners. [Learn more](#)

 Build Your Own By Google Build your own connectors.	 Facebook Ads By Supermetrics #1 connector for Facebook Ads. Free 14 day trial. Trusted by 700k+ marketers.	 Rubii By Rubii Once you set up a custom report in Rubii, your reports will be available to select in the drop down below.
 Digital Opptur: Kobler Data By Digital Opptur AS	 Line Ads By Supermetrics	 Streamlike Analytics By Mediatech

For this lab, you will work on [CustomerLoyaltyProgram.csv](#), which you need to download to your computer first.

You will use the **File Upload** connector to upload the data to Looker Studio to create the data source.

2. In the **Search** box, type *file upload*, then click on the **File Upload** connector.

Search

file upload



Google Connectors (1 of 23)

Connectors built and supported by Looker Studio [Learn more](#)

**File Upload**

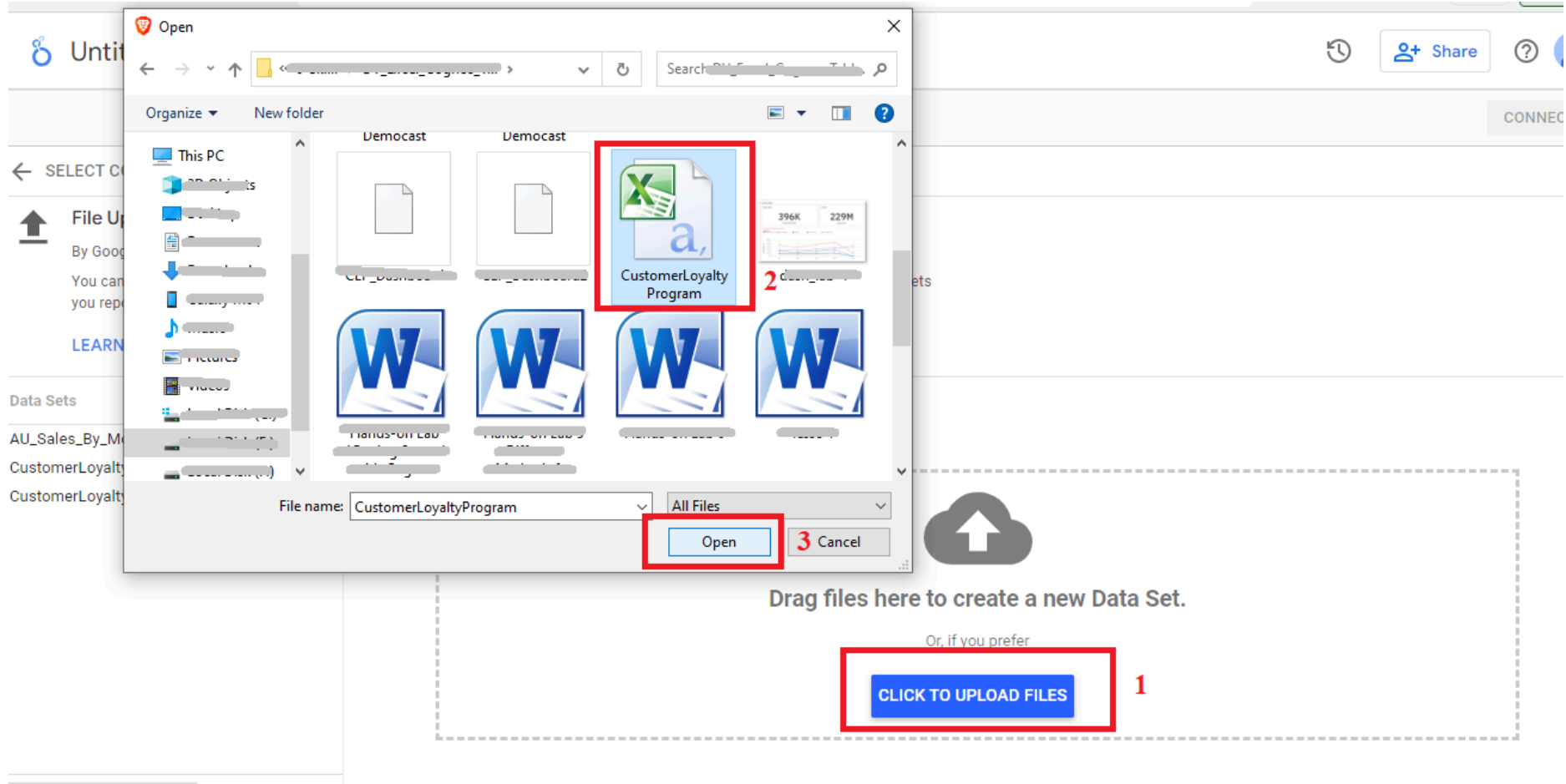
By Google

Connect to CSV (comma-separated values) files.

Partner Connectors (0 of 839)

Connectors built and supported by Looker Studio partners. [Learn more](#)

3. Click the **CLICK TO UPLOAD FILES** button, select the *CustomerLoyaltyProgram.csv* file and click **Open**.



4. Once the data is uploaded, click **CONNECT**.

CONNECT

← SELECT CONNECTOR

 File Upload

By Google

You can bring data into Looker Studio from almost any source by uploading CSV (comma-separated values) files. File upload lets you report on data not supported by a specific connector.

[LEARN MORE](#) [REPORT AN ISSUE](#)

Data Sets 

- CustomerLoyaltyProgram.csv
- AU_Sales_By_Model.xlsx
- CustomerLoyaltyProgram.csv
- CustomerLoyaltyProgram.csv

CustomerLoyaltyProgram.csv

TOTAL FILE SIZE	NUMBER OF FILES	CREATION DATE	LAST MODIFIED DATE
17 MB (18% of 100MB used)	1	9/17/23 4:58 PM	9/17/23 4:58 PM

[VIEW FILES IN CLOUD](#)

[ADD FILES](#) Files must contain the same schema. [Learn More](#)

File name	Uploaded at	Size	Status
CustomerLoyaltyProgram.csv	9/17/23 4:58 PM	17 MB	● Uploaded

Here you can see the contents of the uploaded data source. On this page you can verify or modify the data type of each data attribute, modify the default aggregation, include the description for fields, and add new fields and parameters as well.

5. To start creating the report, click **CREATE REPORT**.

CustomerLoyaltyProgram.csv

Scope: [Reusable](#) | Data credentials: [P P](#) | Data freshness: [12 hours](#) | Community visualizations access: [On](#) | Field editing in reports: [On](#) **Click here**

[CREATE REPORT](#) [EXPLORE](#)

[← EDIT CONNECTION](#) | [FILTER BY EMAIL](#)

4 [+ ADD A FIELD](#) 5 [+ ADD A PARAMETER](#)

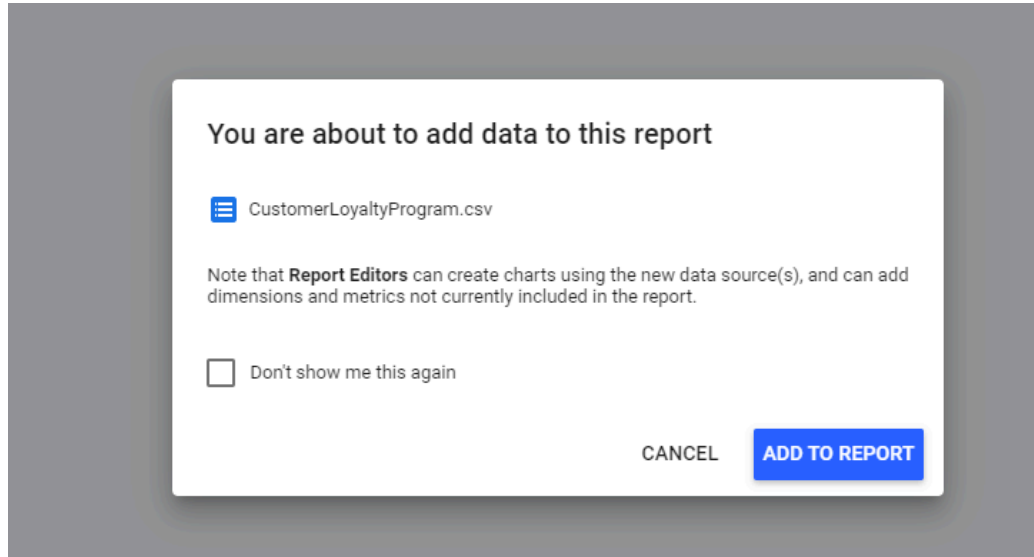
Field ↓	Type ↓	1	Default Aggregation ↓	2	Description ↓	3
DIMENSIONS (28)						
City	ABC	Text	▼	None		
Count	123	Number	▼	Sum	▼	
Country	🌐	Country	▼	None		
Coupon Response	ABC	Text	▼	None		
Customer Lifetime Value	123	Number	▼	Sum	▼	
Customer Name	ABC	Text	▼	None		
Education	ABC	Text	▼	None		
First Name	ABC	Text	▼	None		
Gender	ABC	Text	▼	None		
Income	123	Number	▼	Sum	▼	
Last Name	ABC	Text	▼	None		
Latitude	123	Number	▼	Sum	▼	

1.1

[REFRESH FIELDS](#)

29 / 29 Fields

6. In the pop-up dialog box, click **ADD TO REPORT**.



The **Report Editor** tool will open.

9. To give yourself more screen space and expand the canvas window, you can close the **Data** and **Properties** panes on the right side of the page.

Simple Dashboard

File Edit View Insert Page Arrange Resource Help

Reset Share View ?

Pause updates

Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Data

Search

CustomerLoyaltyProgram.csv

- ABC City
- 123 Count
- ABC Country
- ABC Coupon Response
- 123 Customer Lifetime Value
- ABC Customer Name
- ABC Education
- ABC First Name
- ABC Gender
- 123 Income
- ABC Last Name
- 123 Latitude
- ABC Location Code
- 123 Longitude
- 123 Loyalty Count

+ Add a field

+ Add a parameter

+ Add Data

Data

Properties

NOTE: To work on data in Excel format, upload the .xls file to your computer, and use the 'Google Sheets' connector to create the data source.

Task 2: Use Report Editor

Let's see what tools are available in the Report Editor.

Simple Dashboard

File Edit View Insert Page Arrange Resource Help

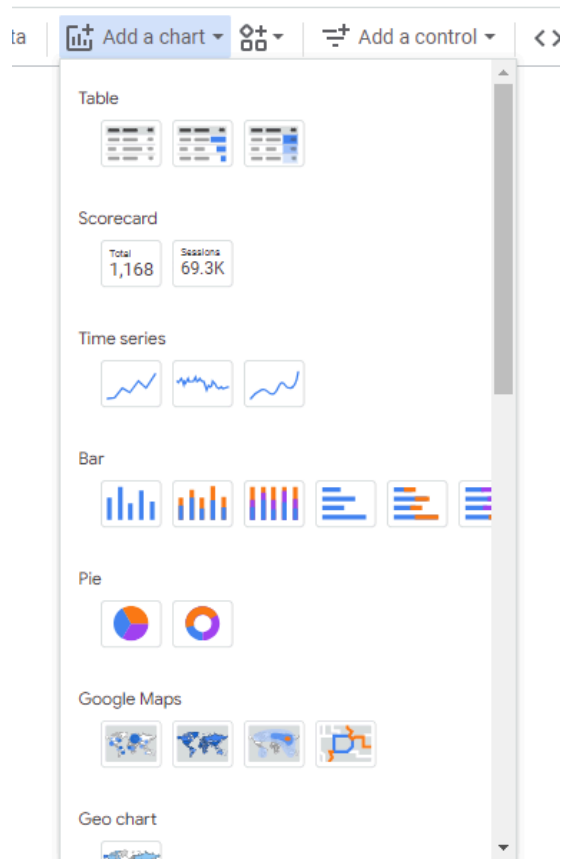
Reset Share View

Undo Redo Select Zoom Add page Add data Add a chart Add a control Pause updates

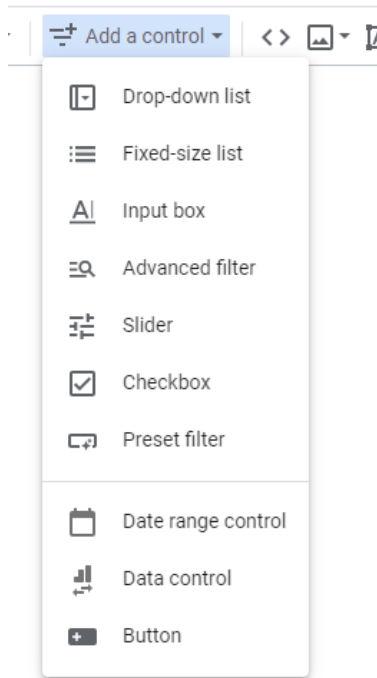
Data Properties

10

1. To add a new chart, click **Add a chart**. Looker Studio provides a variety of charts to be used for creating visualizations such as tables, scorecards, time series charts, bar charts, line charts, pie charts, and maps to name but a few.

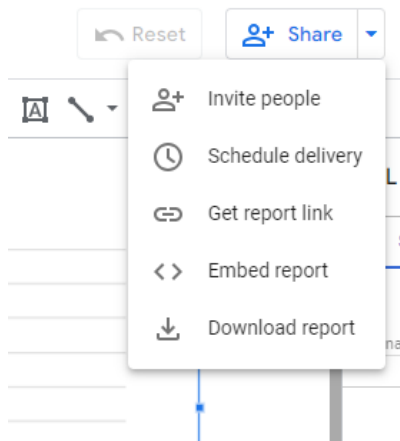


2. Scroll down to see all the options. To include data, click **Add data**, then close the **Add data to report** window.
3. Click **Add a control**. Controls are used to make your visuals interactive. Looker Studio provides several control options including sliders, filters, checklists, drop-down lists, and buttons.



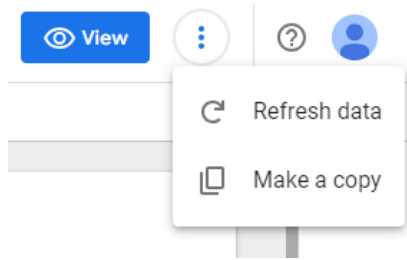
Controls enable you to adjust the data shown in report components by filtering or modifying it. They serve as a means to collect user input and incorporate it into calculated fields.

4. Use the icons to the right of **Add a control** to insert components other than charts and controls to your dashboard or report. These include URLs, images, textboxes, and lines and shapes. To access the **Theme and layout** option, if it is hidden, click the elipsis button (vertical three dots).
5. The **Share** button lets you share your report with others.

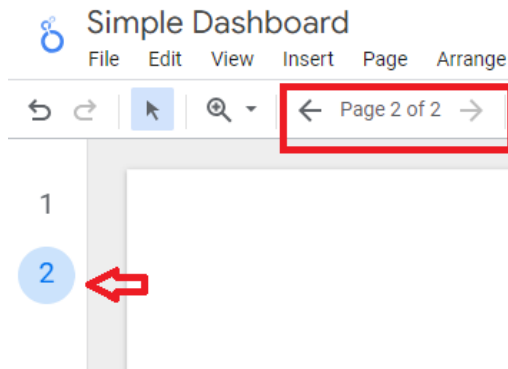


You can invite your colleagues to work on your dashboard with you, you can also get the link or embedded code, and you can download the report. You also have the option to schedule the delivery time of your report.

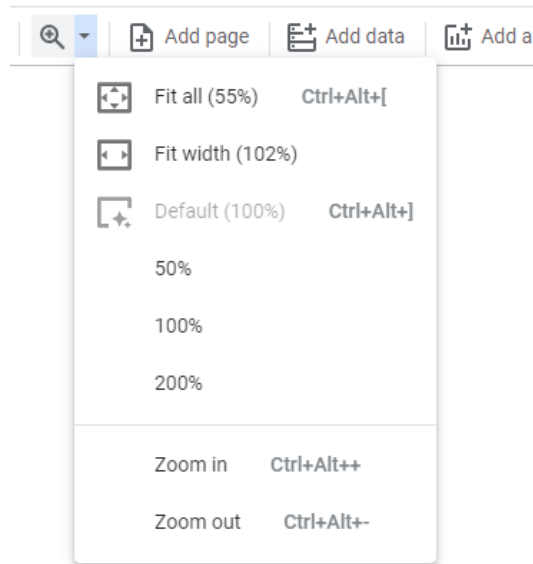
6. If you prefer not to make edits to the report and simply want to see how it appears in read-only mode, click **View**. You can click **Pause updates** to pause the data updates for the live data, if used, and you can refresh or make a copy of the data by clicking on the elipsis button (three vertical dots) here.



7. Click **Add page** to add more pages to your report. You can easily switch amongst pages using the left navigation bar or the arrows in the toolbar.



8. Looker Studio provides several options to zoom in and out, such as **Fit all**, **Fit width**, and various percentage values.



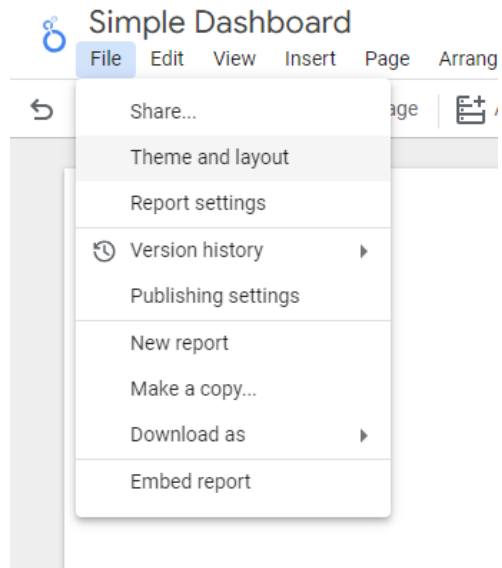
9. Use the **Undo** and **Redo** buttons to fix mistakes or misclicks.

10. The main work area at position 10 is the **canvas** where you add and layout all your visualizations.

Exercise 4: Access Report Themes and Layouts

Unlike Cognos Analytics, Looker Studio gives you the flexibility to place the visuals where you like to while you prepare the report or dashboard. So you don't have to select a fixed dashboard template, as you do in Cognos Analytics. However, Looker Studio does have some inbuilt themes with different color and font combinations for you to choose from.

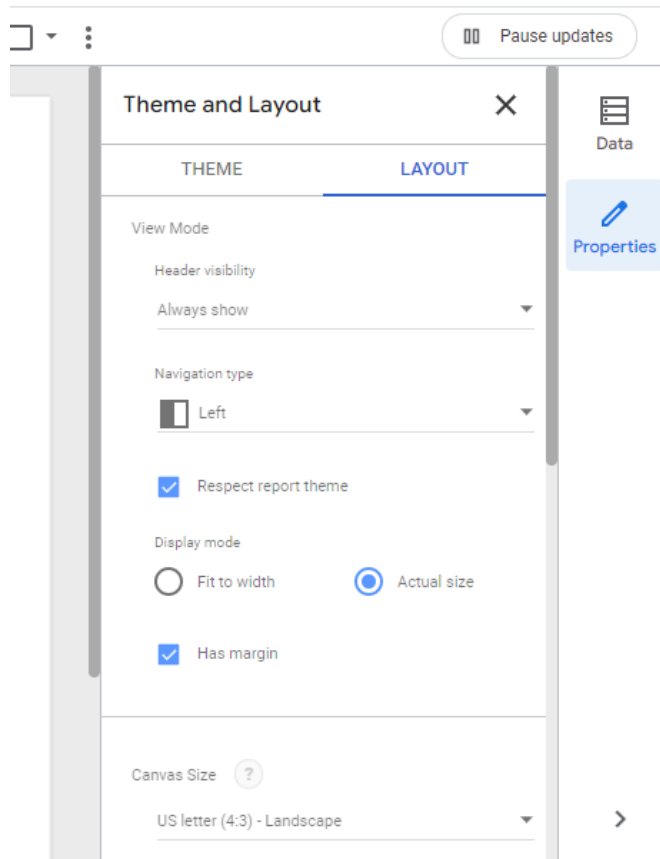
1. To access the *Theme and layout* menu, either click **File** in the main menu, then click **Theme and layout**, or in the toolbar, click **Theme and layout**. If it's hidden, click the elipsis button (...) to show it.



2. Use the **THEME** tab to modify the default theme or select one of the predefined themes for your report.

The screenshot shows the 'Simple Dashboard' application. At the top, there is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. To the right of the menu are buttons for 'Reset', 'Share', 'View', and a user profile icon. Below the menu is a toolbar with icons for undo, redo, select, zoom, add page, add data, add chart, add control, and a 'Pause updates' button. The main canvas is a large white area. On the right side, there is a 'Theme and Layout' panel. This panel has two tabs: 'THEME' and 'LAYOUT'. The 'THEME' tab is currently selected, showing 'Current Theme Default' and a 'Customize' button. Below this, there are two theme preview cards: 'Default' and 'Edge'. The 'Default' card shows a dashboard layout with text, a bar chart, and a pie chart. The 'Edge' card shows a similar layout with a dark theme. At the bottom of the panel is a button labeled 'Extract theme from image'. To the right of the panel is a 'Data' icon and a 'Properties' button.

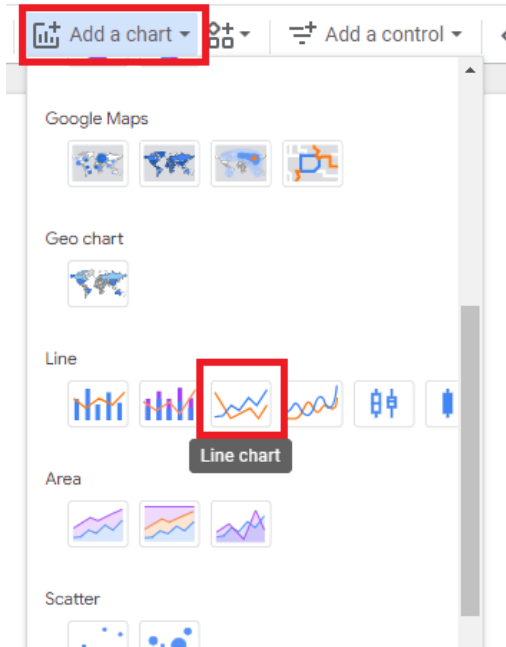
3. Use the **LAYOUT** tab to change the layout of your canvas, such as the type of navigation, canvas size, and grid settings.



Exercise 5: Create a Simple Dashboard Report

Let's create a simple dashboard on **Product Line Performance by Year**

1. Click **Add a chart** and select the simple **Line chart**.



2. Click on the canvas where you want it to be positioned. You can move it anywhere on the canvas later by simply clicking and dragging it to a new position. Looker Studio automatically includes data to create the chart based on the data source.

For your *Product Line Performance by Year* visualization, you place the data as you want it to be displayed. The requirement is to create a line chart for the quantity sold per order year and have separate lines displayed for each product line.

Pause updates

Chart

SETUP STYLE

Data source

CustomerLoyaltyProgram.csv

+ BLEND DATA ?

Date Range Dimension

+ Add dimension

Dimension

123 Order Year

Drill down

Breakdown Dimension

ABC Product Line

Metric

SUM Quantity Sold

Optional metrics

Metric sliders

Data

Properties

3. Click on the line chart in the canvas, and then click **Properties**.
4. Click **Data** to open that pane on the right too.
5. From the data pane, drag **Order Year** to the **Dimension** field to replace **First Name**.

Theme and layout Pause updates

Chart ▼ **Data**

SETUP **STYLE**

Data source
CustomerLoyaltyProgram.csv

+ BLEND DATA ?

Date Range Dimension
+ Add dimension

Dimension
123 Order Year

Drill down ☐

Breakdown Dimension
+ Add dimension

Metric

Search

CustomerLoyaltyProgram.csv

- 123 Income
- ABC Last Name
- 123 Latitude
- ABC Location Code
- 123 Longitude
- 123 Loyalty Count
- 123 Loyalty#
- ABC LoyaltyStatus
- ABC Marital Status
- 123 MonthsAsMember
- 123 Order Year
- ABC Postal code
- ABC Product Line
- ABC Province or State

Data

Properties

6. From the data pane, drag **Quantity Sold** to the **Metric** field. Remove the **Record Count** item.

The screenshot shows a data visualization tool interface. At the top, there is a toolbar with icons for image, text, and shape, followed by a 'Theme and layout' button and a 'Pause updates' button. The main interface is divided into three panes: 'Chart', 'Data', and 'Properties'. The 'Chart' pane is active and shows the 'SETUP' tab. It has two sub-tabs: 'SETUP' and 'STYLE'. Under 'SETUP', there are sections for 'Date Range Dimension' (with an 'Add dimension' button), 'Dimension' (with a dropdown menu showing '123 Order Year'), 'Drill down' (with a toggle switch), and 'Breakdown Dimension' (with an 'Add dimension' button). The 'Metric' section is highlighted with a red rectangle and shows a dropdown menu with 'SUM' and 'Quantity Sold'. The 'Data' pane is on the right and shows a search bar and a list of fields from the 'CustomerLoyaltyProgram.csv' file. The fields are: Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, Marital Status, MonthsAsMember, Order Year, Postal code, Product Line, and Province or State. The 'Properties' pane is on the far right and shows a 'Data' button and a 'Properties' button.

You want to break down the chart by product line, so that it can display a separate line for each product category.

7. From the data pane, drag **Product Line** to the **Breakdown dimension** field.

Chart

Theme and layout

Pause updates

Chart

SETUP

STYLE

Date Range Dimension

+ Add dimension

Dimension

123 Order Year

Drill down

Search product line

Breakdown

Default group

+ ABC Product Line

Metrics

SUM

+ Data

Properties

CustomerLoyaltyProgram.csv

123 Income

ABC Last Name

123 Latitude

ABC Location Code

123 Longitude

123 Loyalty Count

123 Loyalty#

ABC LoyaltyStatus

ABC Marital Status

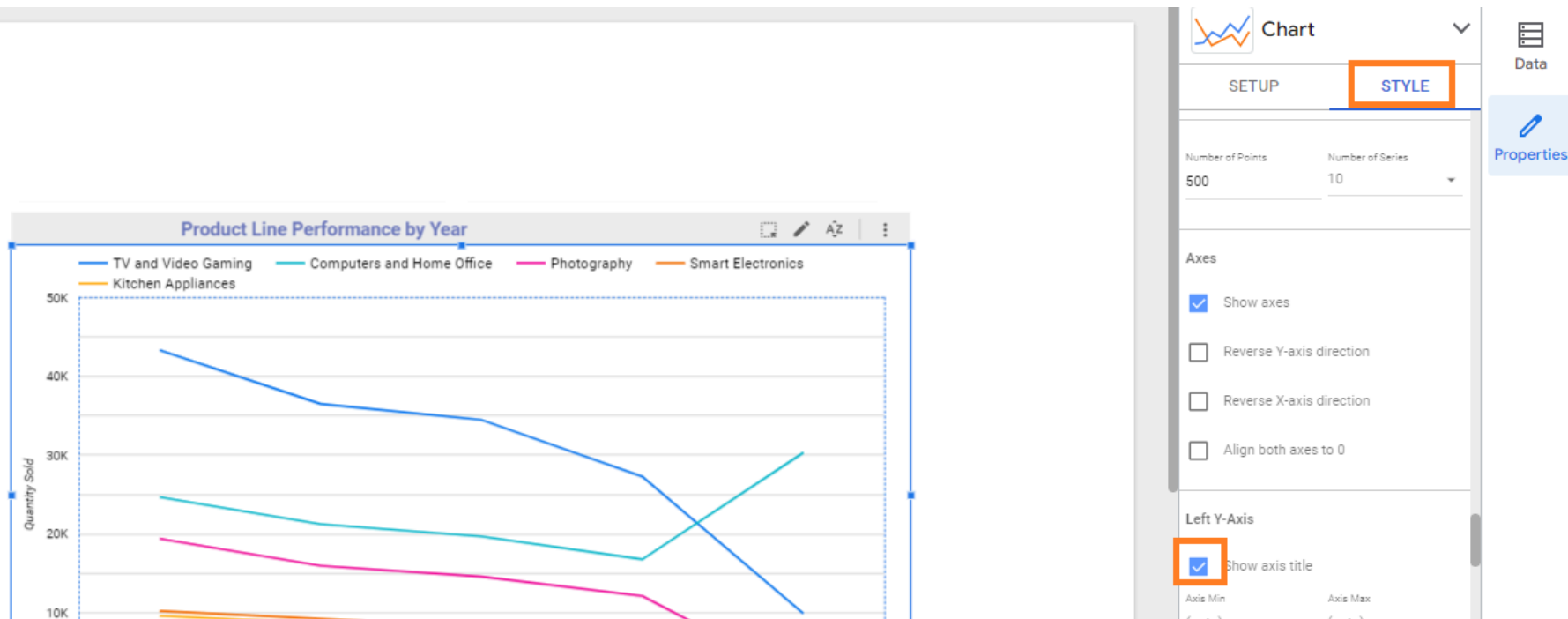
123 MonthsAsMember

123 Order Year

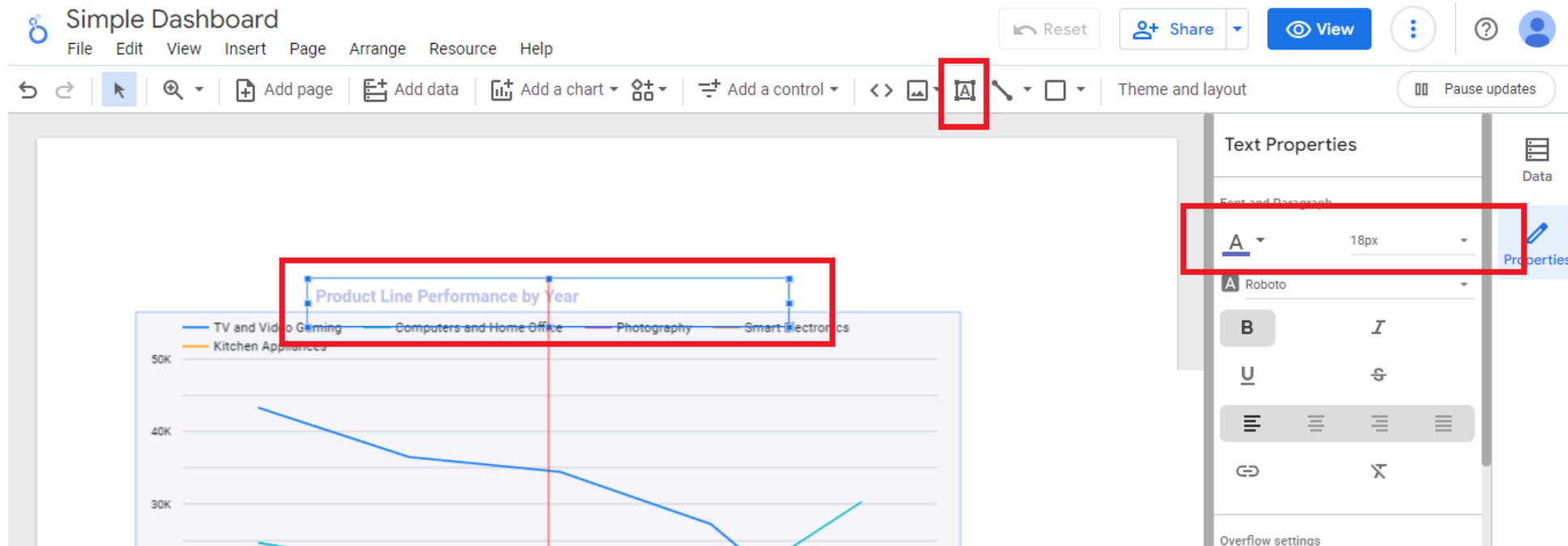
ABC Postal code

ABC Product Line

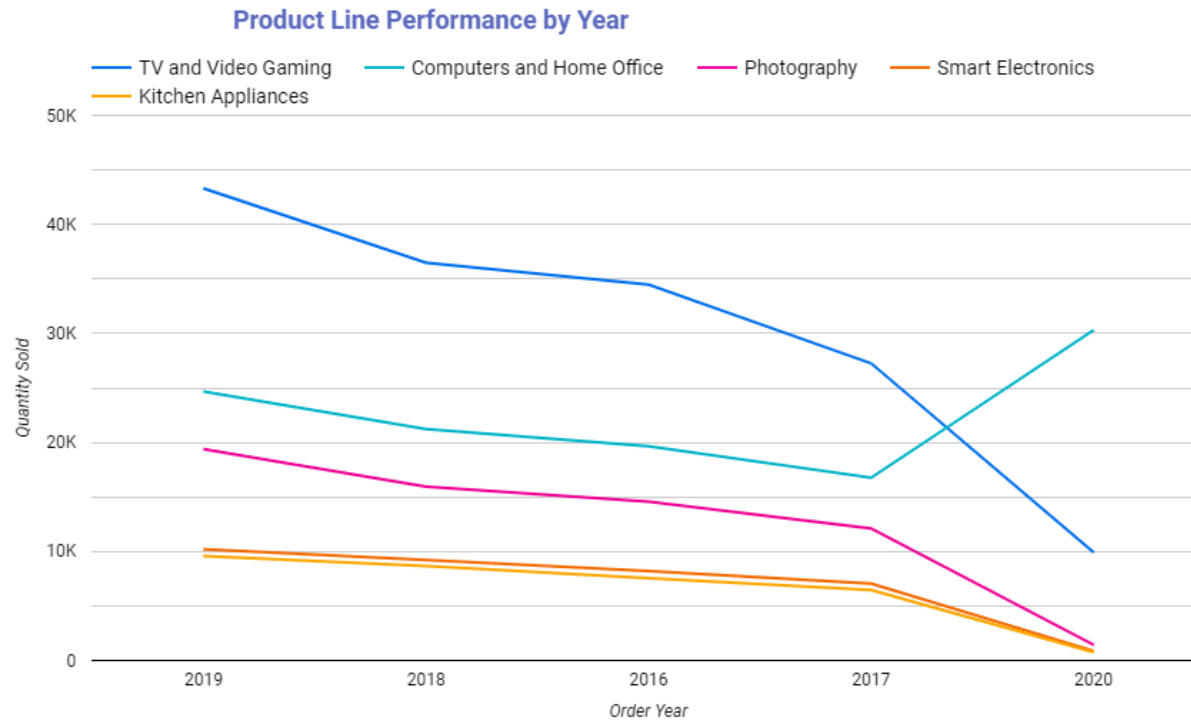
8. To include the x and y axis labels, click the **STYLE** tab in the chart's **Properties** pane, and check the box for **Show axis title** in both the **Left Y-axis** and the **X-axis** sections.



9. Hover over the bottom right corner of the chart till you see the white double-headed arrow, then click and drag to make the chart larger.
10. In the main toolbar, select the **Text** tool and click above the visualization to insert a text box for the chart title. Click in the text box and type the title as *Product Line Performance by Year*.
11. Select the text in the new title and use the **Text Properties** in the right pane to style the text as **24pt, bold, and dark blue**.
12. Drag the text box to align it with the center of the line chart visualization, and drag the chart and the title boxes down the page a bit to make some room at the top for the next visualization.



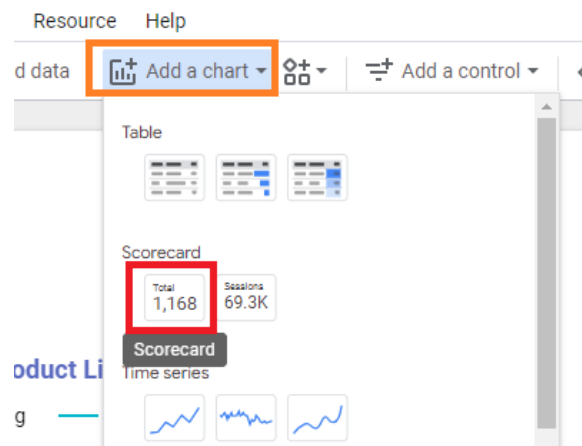
Your line chart should now look similar to the image below.



Now you will include two scorecards to display the *Total Quantity Sold* and *Revenue* above this line chart.

13. In the toolbar, click **Add a chart**, and select **Scorecard**.

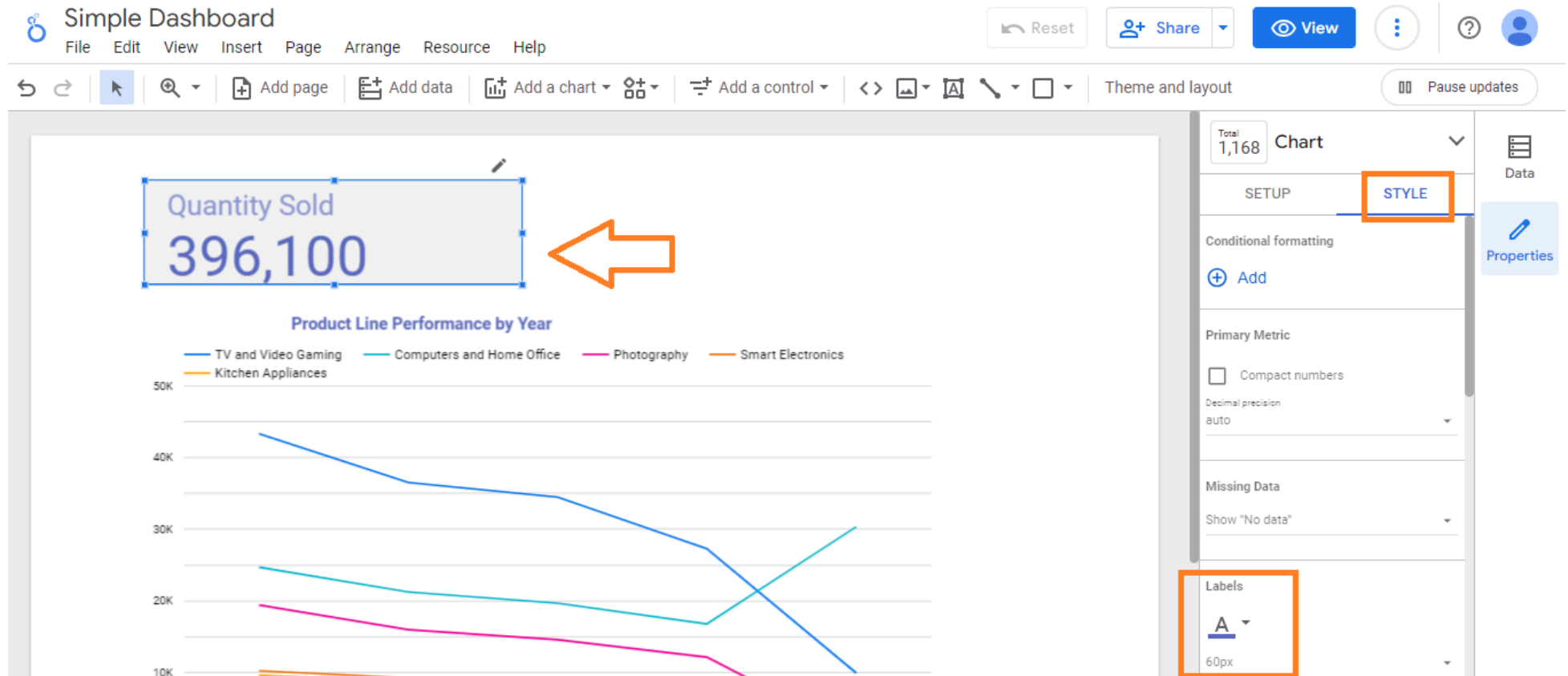
14. Move it above the line chart visualization and to the left side of the canvas.



Looker Studio will automatically pick **Quantity Sold** to be displayed on this scorecard.

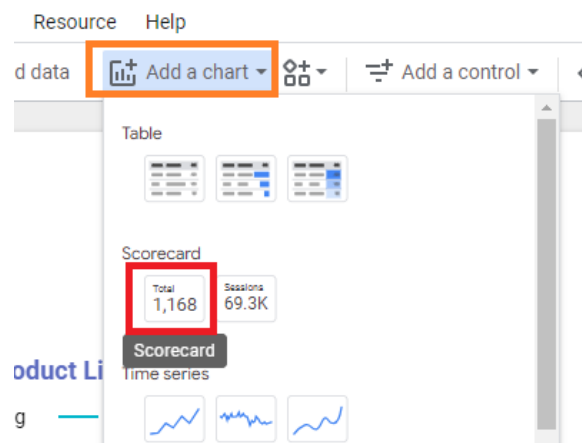
15. You can change the size and position as you like.

16. Use the **STYLE** tab in the scorecard chart's **Properties** pane to change the font size and color to **48pt** and **dark blue**.

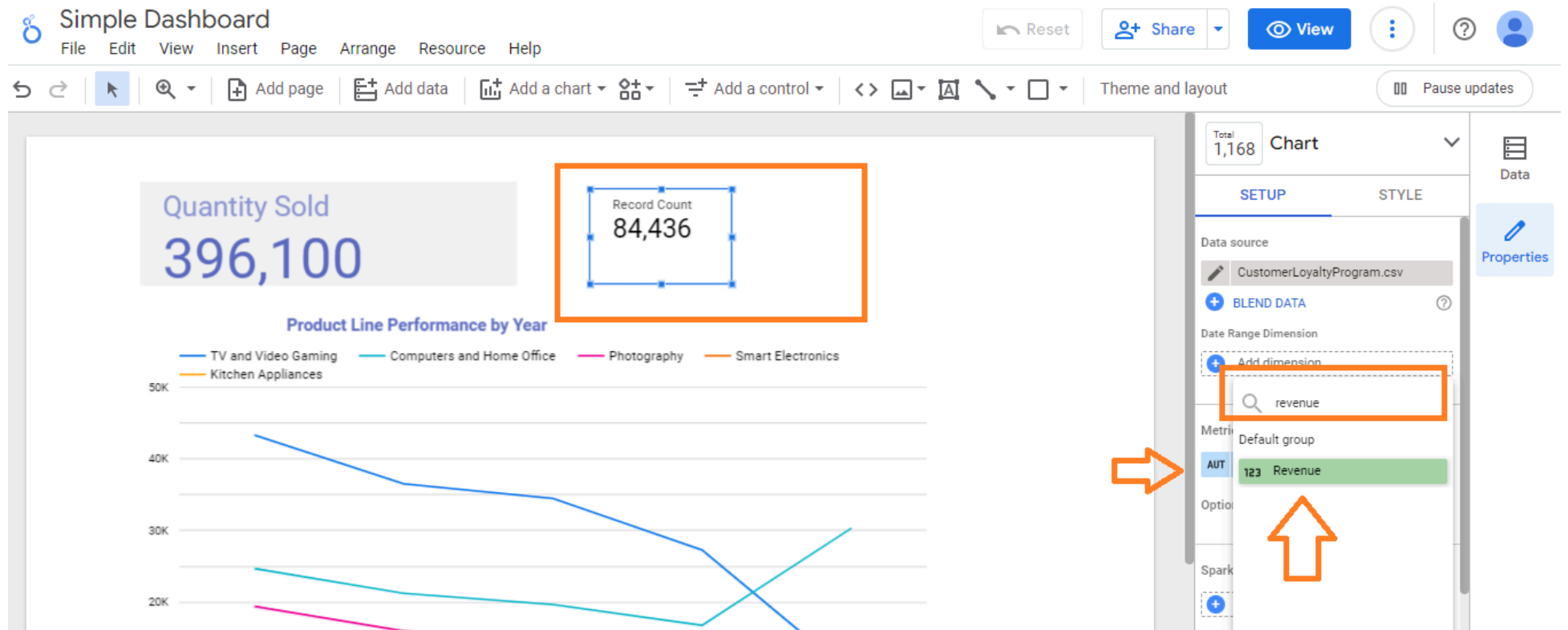


Now you will add the second scorecard chart above the line chart.

17. In the toolbar, click **Add a chart**, and select **Scorecard**.



18. Place it to the right of the **Quantity Sold** scorecard chart.



This time Looker Studio has picked **Record Count** to create this scorecard.

Let's change the metric to show *Revenue* instead.

19. Select the **SETUP** tab in the scorecard chart's **Properties** pane.

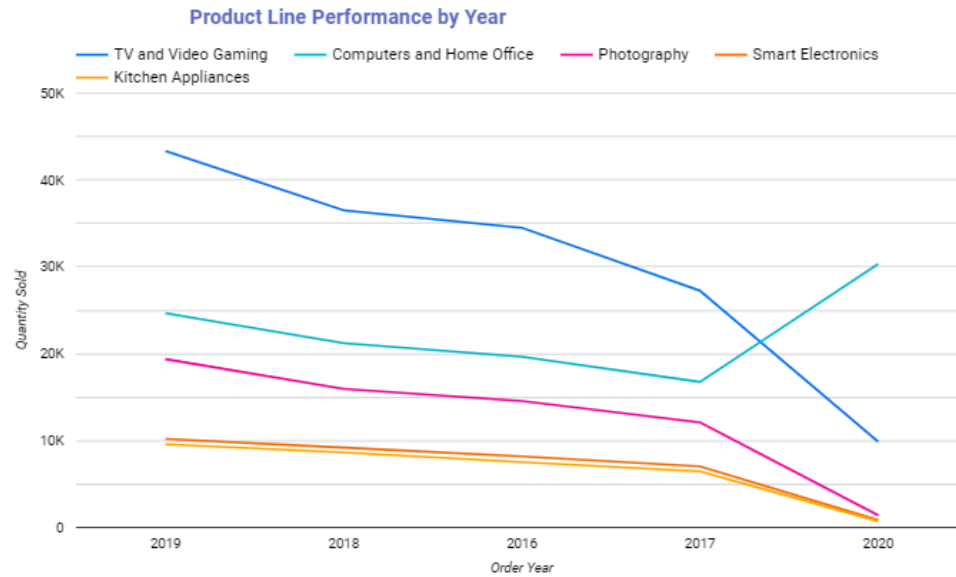
20. From the **Data** pane, drag **Revenue** to the **Metric** field to replace **Record Count**.

21. Use the **STYLE** tab in the scorecard chart's **Properties** pane to change the font size and color to **48pt** and **dark blue** as you did for the previous scorecard chart.

The final version of your first dashboard should appear similar to the image below.

Quantity Sold
396,100

Revenue
228,741,556



Congratulations! You have completed this hands-on lab and you are now ready for the next topic.

For more help, you can refer to the [Tutorial on Looker Studio by Google](#)

Author(s)

[Dr. Pooja](#)